# PARISHRAM 2.0

## CLASS 12TH 2026

Inorganic chemistry

# d & f-Block

CHEMISTRY | LECTURE-1

BY SANYA NAYAR MA'AM



# TOPICS TO BE COVERED

1. General Introduction
2. The Transition Elements (d-Block)




ROAR
Tiger hai tu Tiger! Roar

# General Introduction

**d-Block Elements** : Elements in which the last $e^-$ enters the **d-subshell.**

d-Subshell → 5 orbitals




**d-Block** : 10 Groups are present from **Group 3 to Group 12**

| Subshell | orbitals |
| --- | --- |
| s | 1 |
| p | 3 |
| d | 5 |
| f | 7 |

1 orbital can have a maximum of 2 electrons

# General Introduction

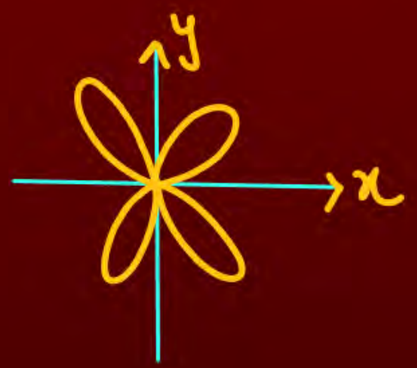
S → Spherical

P → Dumbbell

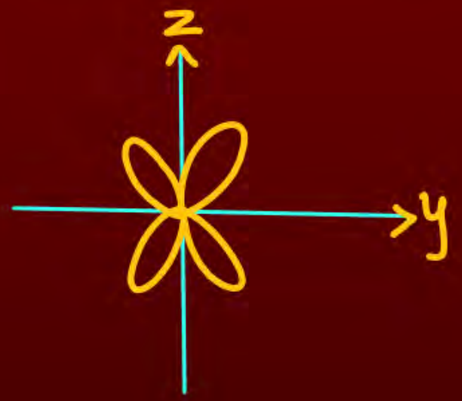
## Shape of d-orbitals

**Double dumbbell shaped except $d_{z^2}$**

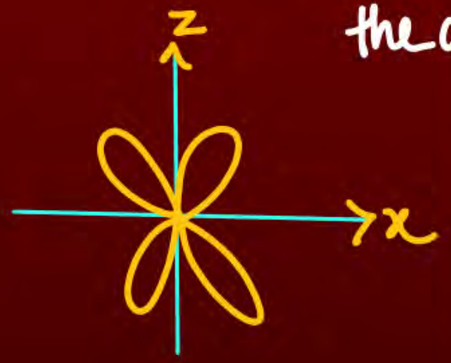
$d_{xy}$




$d_{yz}$

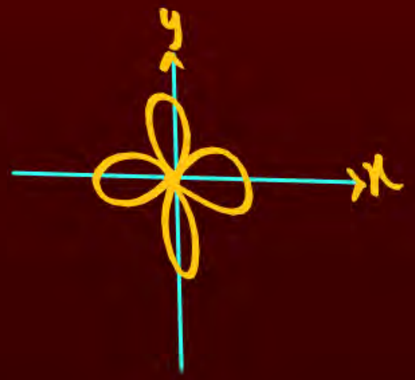



$d_{xz}$

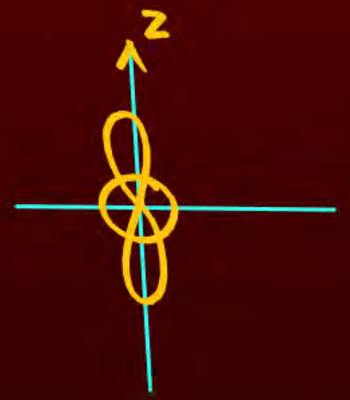



lie in b/w the axes

$d_{x^2-y^2}$




$d_{z^2}$

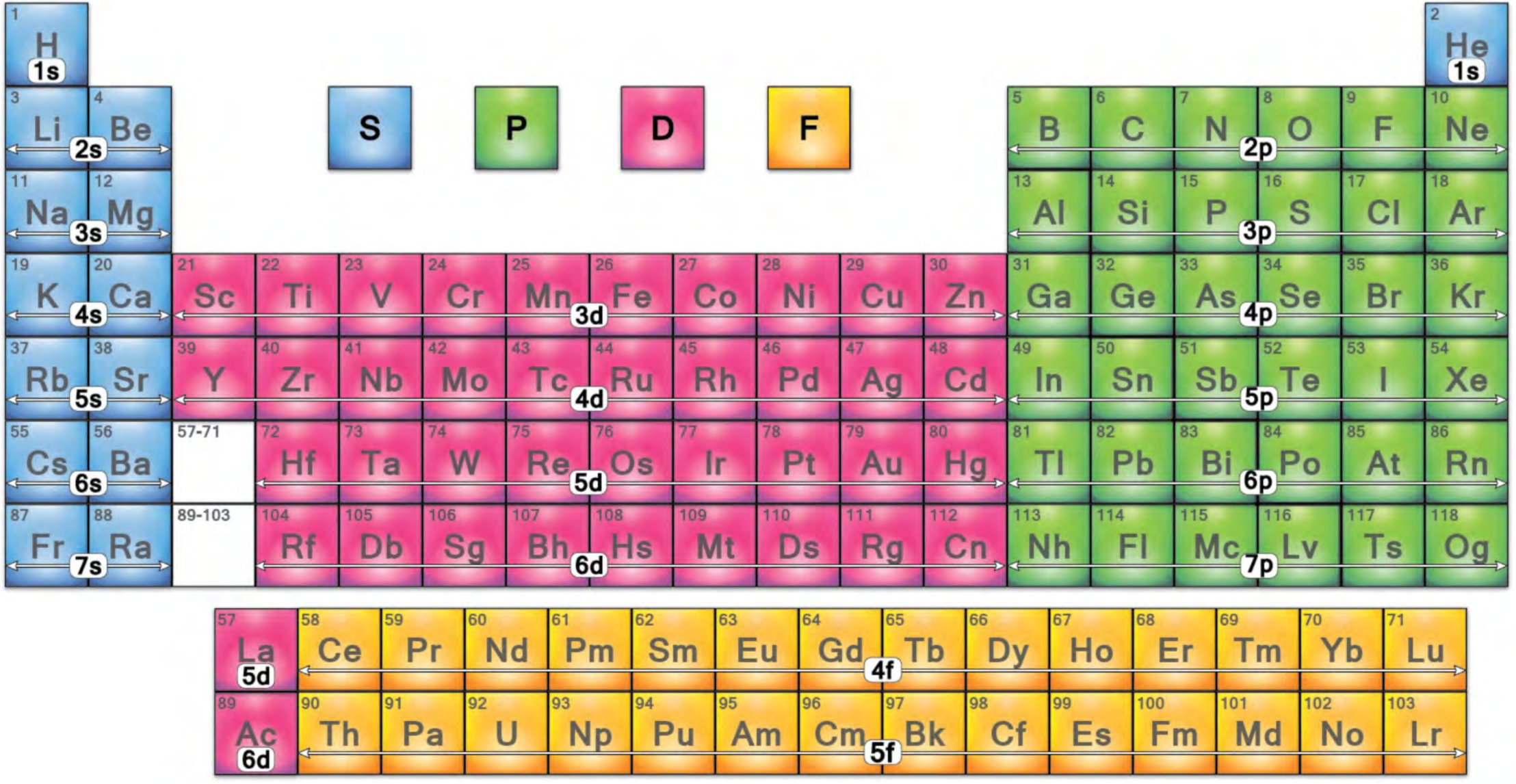



lie on the axes

# General Introduction

**d-Block Elements : 3rd Group – 12th Group**

- d-orbitals are progressively filled in each of the four long periods
- Filled from 4th period
- Transition Elements
- Metals


S
P
D
F
1 H 1s
2 He 1s
3 Li 4 Be 2s
5 B 6 C 7 N 8 O 9 F 10 Ne 2p
11 Na 12 Mg 3s
13 Al 14 Si 15 P 16 S 17 Cl 18 Ar 3p
19 K 20 Ca 4s
21 Sc 22 Ti 23 V 24 Cr 25 Mn 26 Fe 27 Co 28 Ni 29 Cu 30 Zn 3d
31 Ga 32 Ge 33 As 34 Se 35 Br 36 Kr 4p
37 Rb 38 Sr 5s
39 Y 40 Zr 41 Nb 42 Mo 43 Tc 44 Ru 45 Rh 46 Pd 47 Ag 48 Cd 4d
49 In 50 Sn 51 Sb 52 Te 53 I 54 Xe 5p
55 Cs 56 Ba 6s
57-71
72 Hf 73 Ta 74 W 75 Re 76 Os 77 Ir 78 Pt 79 Au 80 Hg 5d
81 Tl 82 Pb 83 Bi 84 Po 85 At 86 Rn 6p
87 Fr 88 Ra 7s
89-103
104 Rf 105 Db 106 Sg 107 Bh 108 Hs 109 Mt 110 Ds 111 Rg 112 Cn 6d
113 Nh 114 Fl 115 Mc 116 Lv 117 Ts 118 Og 7p
57 La 5d
58 Ce 59 Pr 60 Nd 61 Pm 62 Sm 63 Eu 64 Gd 65 Tb 66 Dy 67 Ho 68 Er 69 Tm 70 Yb 71 Lu 4f
89 Ac 6d
90 Th 91 Pa 92 U 93 Np 94 Pu 95 Am 96 Cm 97 Bk 98 Cf 99 Es 100 Fm 101 Md 102 No 103 Lr 5f

# Transition Elements

**Are those elements in which d-orbitals are incompletely filled in either Ground state or in any of its oxidation states.**

Exception

**Group-12**

(n=4) Zn

(n=5) Cd

Hg

Cn

General outer electronic configuration

$(n-1)d^{10}\ ns^2$

**Ground State**

$3d^{10}\ 4s^2$

$4d^{10}\ 5s^2$

$5d^{10}\ 6s^2$

$6d^{10}\ 7s^2$

} Completely filled d-orbital

**Common O.S = +2**

$Zn^{+2}$ $3d^{10}$

$Cd^{+2}$ $4d^{10}$

$Hg^{+2}$ $5d^{10}$

$Cn^{+2}$ $6d^{10}$

} Completely filled d-orbital

# Transition Elements

**Group 12** elements are d-block elements as last electron enters into d-subshell but they are not considered as **transition elements** because they have completely filled d-orbitals in ground state as well as in common oxidation state.

Pseudo Transition Elements.

Also their chemistry studied along with transition elements because they are the last members of d-block.

# QUESTION

→ incompletely filled d-orbital

**How can we say that Sc (Z = 21) is a transition element while Zn (Z = 30) is not?**



$Sc\ [Z=21] : [Ar]^{18}\ 3d^1\ 4s^2$



Incompletely filled d-orbital
∴ Sc is a T.E

$Zn\ [Z=30] : [Ar]^{18}\ 3d^{10}\ 4s^2$



→ completely filled d-orbital

$Zn^{+2} : [Ar]^{18}\ 3d^{10}$



→ completely filled d-orbital
∴ Zn is not a T.E

# QUESTION



**Silver atom has completely filled d-orbitals ($4d^{10}$) in its ground state. How can we say that it is a transition element?**

(G.S) $Ag : 4d^{10}5s^{1}$

$Ag^{+1} : 4d^{10}$

$Ag^{+2} : 4d^{9}$



↝ we can see incompletely filled d-orbital in O.S

Hence Ag is a T.E

# QUESTION

**Copper atom has completely filled d-orbitals ($3d^{10}$) in its ground state. How can we say that it is a transition element?**



(G.S) Cu : $3d^{10}4s^1$

$Cu^{+1}$ : $3d^{10}$

$Cu^{+2}$ : $3d^9$ ⇝ incompletely filled d-orbital

# Inner Transition Elements

**f-Block elements :**

- Are those in which last $e^-$ enters into f-subshell.
- Filling of 4f and 5f orbitals take place.
- Formal members of groups 3 but taken out & placed below the periodic table in two long periods.
- Inner Transition Elements
- Metals

# Inner Transition Elements

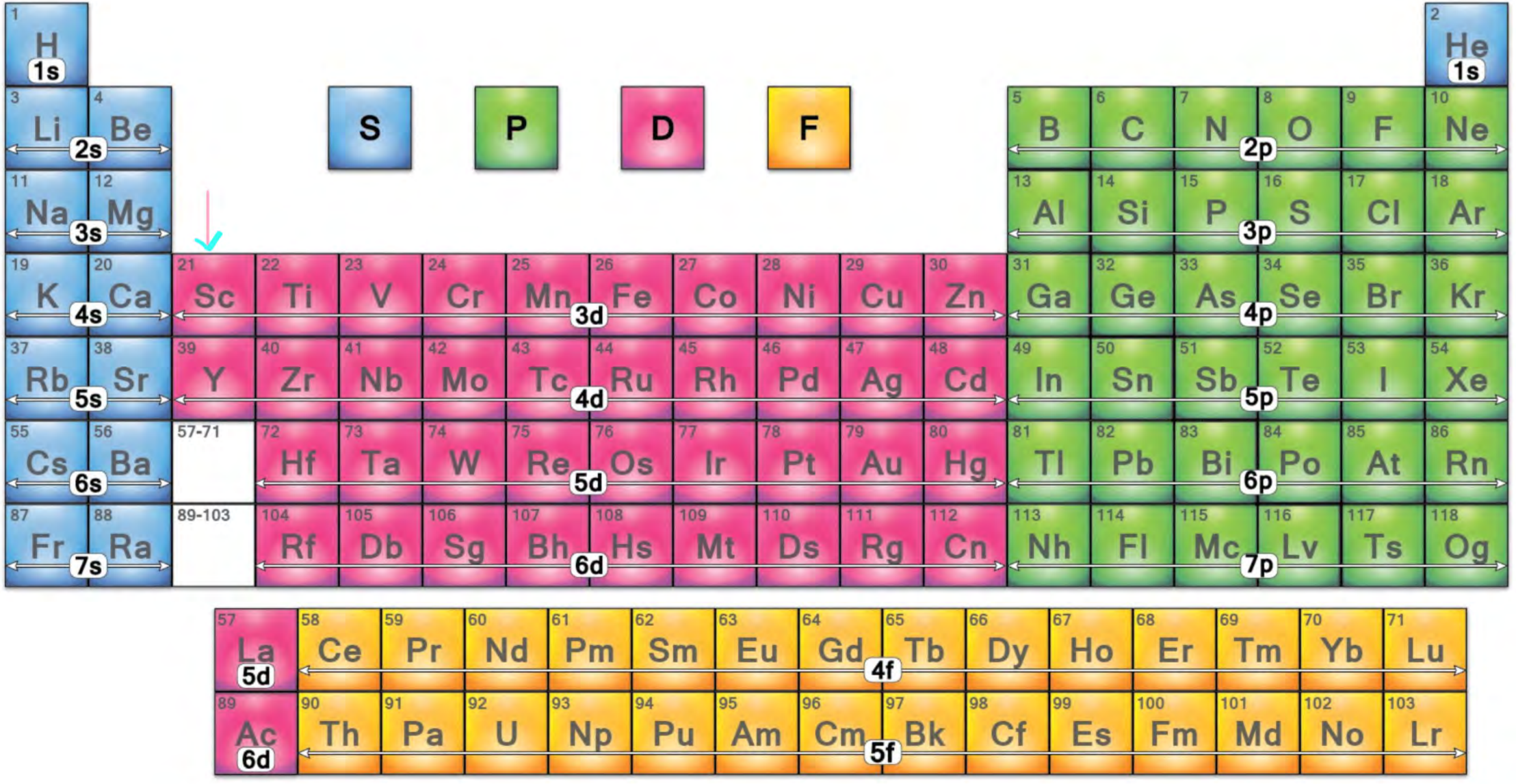
**f-Block**

## Lanthanoids / Lanthanides

- Filling of 4f orbital
- Group-3
- 6th Period
- ${}_{58}Ce - {}_{71}Lu$
- 1st Transition Series

## Actanoids / Actinides

- Filling of 5f orbital
- Group-3
- 7th Period
- ${}_{90}Th - {}_{103}Lr$
- 2nd Transition Series


S
P
D
F
1 H 1s
2 He 1s
3 Li 4 Be 2s
5 B 6 C 7 N 8 O 9 F 10 Ne 2p
11 Na 12 Mg 3s
13 Al 14 Si 15 P 16 S 17 Cl 18 Ar 3p
19 K 20 Ca 4s
21 Sc 22 Ti 23 V 24 Cr 25 Mn 26 Fe 27 Co 28 Ni 29 Cu 30 Zn 3d
31 Ga 32 Ge 33 As 34 Se 35 Br 36 Kr 4p
37 Rb 38 Sr 5s
39 Y 40 Zr 41 Nb 42 Mo 43 Tc 44 Ru 45 Rh 46 Pd 47 Ag 48 Cd 4d
49 In 50 Sn 51 Sb 52 Te 53 I 54 Xe 5p
55 Cs 56 Ba 6s
57-71
72 Hf 73 Ta 74 W 75 Re 76 Os 77 Ir 78 Pt 79 Au 80 Hg 5d
81 Tl 82 Pb 83 Bi 84 Po 85 At 86 Rn 6p
87 Fr 88 Ra 7s
89-103
104 Rf 105 Db 106 Sg 107 Bh 108 Hs 109 Mt 110 Ds 111 Rg 112 Cn 6d
113 Nh 114 Fl 115 Mc 116 Lv 117 Ts 118 Og 7p
57 La 5d
58 Ce 59 Pr 60 Nd 61 Pm 62 Sm 63 Eu 64 Gd 65 Tb 66 Dy 67 Ho 68 Er 69 Tm 70 Yb 71 Lu 4f
89 Ac 6d
90 Th 91 Pa 92 U 93 Np 94 Pu 95 Am 96 Cm 97 Bk 98 Cf 99 Es 100 Fm 101 Md 102 No 103 Lr 5f

# The Transition Elements (d-Block)

**The very name 'Transition' given to the elements of d-block is only because of their position between s-and p-block elements.**

## Electronic configurations of the d-block elements:

Pd.

- 4th • **For 3d Series ($_{21}$Sc to $_{30}$Zn):** $3d^{1-10}\ 4s^{1-2}$
- 5th • **For 4d Series ($_{39}$Y to $_{48}$Cd):** $4d^{1-10}\ 5s^{0-2}$
- 6th • **For 5d Series ($_{57}$La to $_{80}$Hg):** $5d^{1-10}\ 6s^{1-2}$
- 7th • **for 6d Series ($_{89}$Ac to $_{112}$Cn):** $6d^{1-10}\ 7s^{1-2}$

**General E.C. is**

[Pd : $4d^{10}\ 5s^{0}$]

# The Transition Elements (d-Block)

**General Electronic configurations of the d-block elements:**

$$(n-1)d^{1-10}\ ns^{0-2}$$

$n = 4$ to $7$

## 3d-series / 1st series / 4th period

| 1st Series | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Sc | Ti | V | Cr | Mn | Fe | Co | Ni | Cu | Zn |
| $Z$ | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| $4s$ | 2 | 2 | 2 | 1 | 2 | 2 | 2 | 2 | 1 | 2 |
| $3d$ | 1 | 2 | 3 | 5 | 5 | 6 | 7 | 8 | 10 | 10 |

## 4d-series / 2nd series / 5th period

Yaari Zara Nibhana Hout TK Rote Rehna Padega Aaj Cal

| 2nd Series | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Y | Zr | Nb | Mo | Tc | Ru | Rh | Pd | Ag | Cd |
| $Z$ | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| $5s$ | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 2 |
| $4d$ | 1 | 2 | 4 | 5 | 6 | 7 | 8 | 10 | 10 | 10 |

# Homework

→ Read NCERT + Q's.





# THANK YOU